Tire - Airborne Noise Source Level Chassis Dynamometer Evaluation Procedure

1 Scope

Note: Nothing in the standard supersedes applicable laws and regulations unless specific exemption has been obtained.

1.1 Purpose. The purpose of this test procedure is to measure the tire airborne noise source level when using a tire noise testing trailer or an equivalent fixture on a chassis dynamometer (dyno).

The source level can be described by either the near-field sound intensity as shown in test procedure GMW14239 or the far-field sound power in accordance with ANSI S12.35-1990.

1.2 Foreword. The use of the dynamometer establishes all weather or all season capability and improves data repeatability and work efficiency.

This procedure requires a two-roll chassis dynamometer. Since the dyno roll surface has a periodically random excitation and it cannot be perfectly cast from the real road surface, the actual road results and dynamometer results differ in terms of absolute values. In general, the relative discrepancies between two tires measured on the dynamometer are highly consistent with those on the road.

1.3 Applicability. This test procedure applies to multipurpose passenger vehicle and light-duty truck tires.

2 References

Note: Only the latest approved standards are applicable unless otherwise specified.

2.1 External Standards/Specifications.

ANSI S12.35-1990

2.2 GM Standards/Specifications.

GMW14172 GMW14239

2.3 Additional References. None.

3 Resources

3.1 Facilities. The dynamometer tire noise tests will be performed on a hemi-anechoic chassis dynamometer. Refer to Appendix A for the recommended facilities used in NA.

3.2 Equipment. Utilize the similar equipment as specified in road tire noise testing procedure GMW14239, or utilize the equivalently performed LMS recording system, which is recommended in NA. See Appendix A for a reference.

Utilize a pair of phase-matched microphones for intensity measurements whose requirements are specified in 4.1.8.

3.3 Test Vehicle/Test Piece. Two (2) identical tires (or two different tires with equivalent diameters) are required for one test; one tire on each side of the trailer. Only one tire is tested and the other tire is used to retain the same axle height. Therefore, the tested tire shall be marked or specified before the test.

3.4 Test Time.

Calendar time:	2 to 3 days
Test hours:	4-8 hours (relying on sample number) 8 hours (roll cap changeover time 1 to 2 hours (uploading/processing data 1 to 2 hours (down loading/plotting)
Coordination hours:	1 to 3 hours

3.5 Test Required Information. Tire manufacturer, tire trade name, tire type, construction number, inflation pressure, test conditions (speeds and roads), and GM vehicle program name or competitors' vehicle names (if their tires are tested), etc. shall be given for data management. In particular, the construction number of a tested tire shall be provided if the tire is not a production tire.

3.6 Personnel/Skills. To complete this test, at least one technician and/or one engineer skilled in noise and vibration data acquisition, analysis, equipment usage, and test setup is required. A skilled dynamometer operator is also required.

4 Procedure

4.1 Preparation.

4.1.1 Road Shell Installation (Appendix A1.1).

4.1.2 Trailer Installation (Appendix A1.2).

4.1.3 Removal of Stones. Any stones embedded inside the treads shall be examined and removed for safety concern as well as shell surface protection.

4.1.4 Dynamometer Operation. Prior to normal dynamometer operation or equipment installation, a manual push on both left and right rolls or low-speed (3 to 6 kph) automatic control shall be conducted to align the left and right trailer tires along the axle direction.

The dynamometer is recommended to operate only on its one roll to drive one tire if both dynamometer rolls can be independently operated.

4.1.5 Tire Warm-up. Test tires shall be run to bring the tires up to their working temperature, equal to or more than 29°C (85°F), before the results are recorded. A non-touch infrared gun meter can be used to monitor the rolling tire's temperature.

4.1.6 Microphone Positions.

4.1.6.1 Sound Intensity Measurement. Lower the intensity probe fixture as shown in Figures E2 through E4 and adjust it so that the centerline of the two microphones is 70 mm above the shell surface, 100 mm outboard away from the sidewall of the test tire.

The front and rear microphone diaphragms shall be in line with the leading edge and the trailing edge of the tire contact patch, respectively, as shown in Figure E4. Because the tire patch is usually in a convex shape, the exact leading edge and trailing edge shall be measured at the centerline of the tire.

4.1.6.2 Sound Power Measurement. The sound power measurement standard, ANSI S12.35-1990, provides the far-field sound power measurement procedure in a hemi-anechoic room. Several sound pressure responses required for power calculation are measured around the hemisphere that surrounds the rotating tire at the center, as seen in Figure E5. In the test, the radius of 1.2 m is chosen to set the hemisphere and to reach the far-field effect. A tire radiates sound symmetrically around the tire central plane ($y = 0$ as shown in the coordinates in Figure E5) since its tread pattern is designed to be symmetric. Therefore, ten microphones as proposed in ANSI S12.35-1990 can be reduced to six microphones as proposed in Figure E5.

An example of placing six microphones with respect to the tire and trailer is shown in Figure E6 where each microphone position is highlighted. Also, a sheet of at least 35 mm thick foam or equivalently high absorption materials shall be placed against the trailer side to avoid the reflection.

4.1.7 Microphone Nose Cones. Prior to intensity tests, the microphone nose cones shall be installed to reduce the low-frequency turbulence effect generated by a rotating tire.

4.1.8 Instrumentation Verification. Verify microphones and signal processing equipment per GMW14172 and GMW14239. The absolute pure tone is used to verify the entire measurement system.

The intensity microphones shall be phase-matched within 0.5 degrees.

4.1.9 Standard Vertical Load. Set a standard vertical test load to each tire. The standard load for tire testing is 70% of the Tire & Rim Association (T&RA) maximum load, or 70% of the European Tyre and Rim Technical Organization (ETRTO) standard load capacity. If the estimated load is more than 750 kg, use 750 kg for the standard load. If the estimated load is less than 450 kg, use 450 kg for the standard load. Because load has been found to have a minimal effect on tire noise, these values have been rounded off to 50 kg increments to assist the operator (since loads must be changed by adding/removing sand bags at the scales).

4.1.10 Other Important Preparation. Comply with the preparation as listed in GMW14239 to set the correct tire pressure, to mount tires and adapters onto the trailer axles, and to install intensity microphones, etc.

4.2 Conditions. Refer to GMW14239.

4.2.1 Environmental Conditions.

4.2.1.1 Ambient Temperature. Test facility should have ambient temperatures between 18 and 29°C (65 and 85°F).

4.2.1.2 Ambient Sound Pressure Level. The ambient sound pressure level prevailing in the dynamometer chamber with the chassis dynamometer energized but not rotating shall not exceed 45 dB(A) in 1/3 octave bands from 200 Hz to 10 kHz.

4.2.2 Test Conditions. Deviations from the requirements of the procedure shall have been agreed upon. Such requirements shall be specified on component drawings, test certificates, reports, etc.

4.3 Instructions. Usually use the Standard Reference Test Tire (SRTT) results to verify whether the overall system is operating correctly or not. Measure the SRTT results at the start of each batch test. Spectra are plotted to compare the SRTT previous results and to examine the data consistency and repeatability.

The following contents provide the specific instructions required by the dynamometer operation; otherwise refer to GMW14239.

4.3.1 Tire/Dynamometer Speed. The tire/dynamometer slow sweeping shall be applied in this procedure to minimize the dynamometer roll periodic resonance and the cap joint periodic impacts.

The sweep speed starts at 10% lower speed and ends at 10% higher speed than a nominal speed. The speed shall linearly run up for 16 seconds. For examples, for 60 kph, sweep the dynamometer speed from 54 to 66 kph within 16 seconds; for 80 kph, sweep the dynamometer speed from 72 to 88 kph within 16 seconds.

4.3.2 Sound Intensity Measurement. Refer to Appendix A3 for NA operation.

4.3.3 Sound Power Measurement. Refer to Appendix A4 for NA operation.

5 Data

5.1 Calculations.

5.1.1 Sound Intensity Calculation. The intensity spectrum calculation is based on two-microphone cross-spectral indirect method, as shown in Appendix B.

For a given microphone location and operating condition, there are two intensity spectra obtained in this procedure, one at the tire leading edge, and the other at the trailing edge. The final result shall be the linear average of both intensity results.

5.1.2 Sound Power Calculation. The sound power calculation is based on ANSI S12.35-1990 far-field hemisphere measurement method. Its formulas are listed in Appendix C.

5.2 Interpretation of Results. For a given tire and running condition, a correction factor between road data acquired per GMW14239 and the results of Section 5.1 can be calculated. This correction can be stored and applied to future dynamometer data on the same type of tires for the same running condition to estimate road tire noise performance.

5.3 Test Documentation. Sound intensity is presented in 32 Hz narrow bands or 1/3 octave bands. Sound power is presented in 1/3 octave bands.

6 Safety

This procedure may involve hazardous materials, operations, and equipment. This method does not propose to address all the safety problems associated with its use. It is the responsibility of the user of the method to establish appropriate safety and health practices and determine the applicability of regulatory limitations prior to use.

7 Notes

7.1 Glossary.

Reference Tire: The Standard Reference Test Tire (SRTT) is commonly used throughout the industry to monitor differences in results and in some cases as a basis to adjust tire measurements to account for differences in test conditions (for example, different road conditions). In tire noise testing, tire noise measurements made with the SRTT are assessed and testing is continued if they are within a specified range.

Sound Intensity: Sound intensity is the acoustic energy flowing through a unit area in a sound field and hence is a vector quantity with an associated direction of propagation. Unit: Watts per squared meter, with a reference value of 10^{-12} used for Sound Intensity Level.

7.2 Acronyms, Abbreviations, and Symbols.

ETRTO European Tyre and Rim Technical Organization

SRTT Standard Reference Test Tire

T&RA Tire & Rim Association

8 Coding System

This standard shall be referenced in other documents, drawings, VTS, CTS, etc. as follows:

Test to GMW14235

9 Release and Revisions

9.1 Release. The standard originated in July 2005, replacing GMN5852. It was first approved by the Global N&V Acoustics and Airborne Noise IDSR Harmonization Team in November 2005. It was first published in November 2005.

9.2 Revisions

Rev	Approval Date	Description (Organization)
A	OCT 2006	Revised. (Global N&V Acoustics and Airborne Noise IDSR Harmonization Team)

Appendix A: Recommended Measurement Facilities, Equipment, and Parameters in NA

A1 Measurement Facilities

The dynamometer hemi-anechoic room, C109 at NVIC - Building 94 on the GM Proving Ground, is capable of providing two epoxy molded shell profiles:

Smooth Road Shell: Duplicating the Smooth Asphalt Interior Noise Road (prior to the 2002 new repaving) on the Milford Proving Ground.

Rough Road Shell: Duplicating the Stud Damaged Concrete Road on the Milford Proving Ground.

A1.1 Road Shell Installation. The specific road shell, as described above, is properly installed in accordance with the test conditions. A 2 to 3 mm thick layer of mastic damping shall be placed on the roll beneath the shell cap joints to minimize the joint abrupt impacts. Shells shall be uniformly bolted onto the roll with a torque force of 47 N-m (35 ft-lbs) on each bolt, making sure that there is no abrupt vertical difference between two adjacent shells. No loose bolts shall be allowed.

Both different shells can be installed together for operation efficiency. Usually, smooth road shells are mounted beneath the right side tire of the trailer and rough road shells beneath the left side tire.

A1.2 Trailer Installation. The GM proprietary trailer as shown in Figures E1 and E2 shall be installed with its tires on the top of the shell surface and supported in the front by a ball-joint fixture that allows for free rotation. A chain shall be utilized to tighten the trailer "decoupler" into the fixture for adding safety. The fixture shall be placed at the centerline of the dyno rolls. The backside of the trailer shall be loosely tightened to the chamber ground with the chains or straps for safety.

A2 Measurement Equipment

Utilize the similar equipment as specified in road tire noise testing procedure GMW14239, or utilize the equivalently performed LMS recording system along with the same type of an intensity probe. It is recommended to use the LMS measurement system for better data management and easier data post-processing.

The following extra equipment and instrumentation, or equivalent, are required to perform this test. See Figure E1 through Figure E4 for photos of the tire noise trailer setup and intensity probe setup. Figures E5 and E6 show the layout of the six microphones surrounding a tire in accordance with ANSI S12.35-1990.

A3 Sound Intensity Measurement Parameters

Refer to GMW14239 for the use of portable tape recorder and portable signal analyzer. Because this test is run in a laboratory environment, the data processing can be carried out directly without external recording as an intermediate step. It is recommended to use the LMS data acquisition measurement system for better data management and easier data post-processing.

All the individual pressure time history in the intensity pair (leading/trailing edge pair or both pairs) shall be simultaneously recorded to be 16 seconds long at a sampling frequency of 32 768 Hz or 16 384 Hz.

The recording trigger shall be synchronized with the start trigger of the dynamometer sweeping. The dynamometer operator and instrumentation operator shall simultaneously press both corresponding start keys.

32 Hz Narrow Band Acquisition Parameters:

- Free run trigger
- 32 Hz resolution
- Maximum frequency: 6400 Hz or higher
- Hanning windows for all the channels
- AC coupled (100 Hz high pass filter)
- 1000 averages (minimum)
- 50% overlapping
- Dynamic range: 72 dB (minimum)

The LMS UPA Tmon programs as shown in Appendix D can be used to process the recorded time history, to automatically set the above default acquisition parameters, and to provide the final results.

A4 Sound Power Measurement Parameters

Follow the time history (16 seconds long) measurement as given in A3 and then calculate the six-microphone

1 Hz resolution for 1/3 octave calculation 1/3 octave SPL spectra based on the parameters set below:

1/3 Octave Band Acquisition Parameters:

- Free run trigger Maximum frequency: 11 000 Hz
- Hanning windows for all the channels

- AC Coupled (100 Hz high pass filter)
- 30 averages (minimum)
- 50% Overlapping
- Dynamic range: 72 dB (minimum)
- “Butterworth” 1/3 octave band filters applied

Or directly measure the six-microphone SPLs based on the similar acquisition setting.

Appendix B: Two-Microphone Cross-Spectral Intensity Method

The intensity can be calculated by the FFT based indirect method [2], that is, from the imaginary part of the cross-power sound pressure spectrum between the two microphones,

$$I = -\frac{1}{\omega \rho \Delta r} \mathrm{Im}(G_{12})$$

Where G_{12} is the cross spectra between two microphones, ω is the angular frequency, ρ is the air density, and Δr is the spacing between two microphones. The mic spacing is usually selected to be16 mm in order to cover the frequency range of interest for tire airborne noise at 400 to 6400 Hz.

The calculation is embedded into the LMS UPA program as shown in Appendix D. However, the same calculation can be easily obtained by any FFT Analyzer or by Excel once the cross-power spectrum is measured.

Appendix C: Far-Field Sound Power Method (ANSI S12.35-1990)

Use the following formula to calculate the total sound power level based on the six-mic SPL dB values:

$$SWL = 10 \bullet \log_{10}\left[\frac{1}{10}\sum_{i=1}^{6} A_i \bullet 10\wedge(SPL_i/10)\right] + 10 \bullet \log_{10}\left(2\pi R^2\right) + 10 \bullet \log_{10}\left(\frac{400}{\rho c}\right)$$

Where:

SPL_i is the measured SPL (dB values) respectively for six microphones

R is the radius of the measurement hemisphere

ρ is the air density; and

c is the speed of sound

Usually:

R=1.2 m

ρ=1.2 kg/m^3 and

c=343 m/s

A_i is the ***i-*** th surface contribution coefficient.

$$A_i = \begin{cases} 1.5 & i = 1 \\ 1.5 & i = 2 \\ 3.0 & i = 3 \\ 1.5 & i = 4 \\ 1.5 & i = 5 \\ 1.0 & i = 6 \end{cases}$$

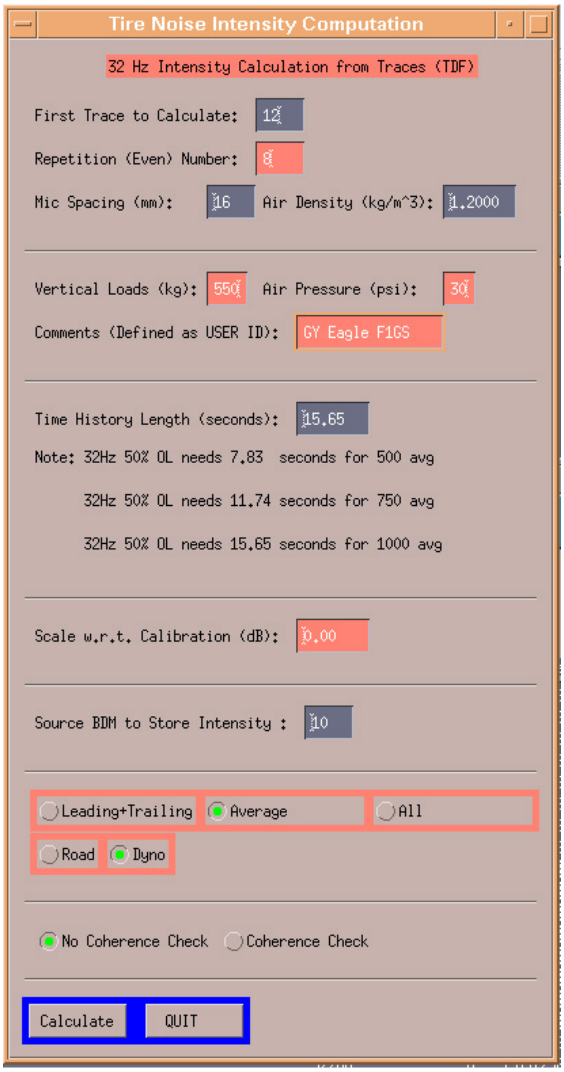
Where p_i^2 is the ***i-*** th measured sound pressure autopower (***pa^2***) respectively from six microphones. The microphone positions are located on a hemisphere with 1.2 meter away from the center of the tire contact patch and given in the table inside Figure E5. An example of placing 6 microphones with respect to the tire and trailer is shown in Figure E6 where each microphone position is highlighted.

Appendix D: LMS UPA Program

Two LMS UPA programs, “tr-int” and “tr-int-oct”, were written to perform the 32 Hz narrow bands and 1/3 octave bands intensity calculation based on the two-microphone cross-spectral indirect method as described in Appendix B. The UPA programs provide the user with the interface as shown on the right side. The programs request the input of the time traces and the relative information. Finally the programs save the calculated intensity values in the specified BDM for display and also save them into the corresponding Test ID (Test ID is the same as their trace Primary ID), which can be retrieved for future display.

The time trace signals are the time history of the intensity probe microphone signals that can be either road inputs or dynamometer inputs.

How to use the right interface is given in details as follows:



First Trace to Calculate: The starting trace number, as an example, starting at 12.

Repetition (Even) Number: The total number of microphone signals as sequentially listed in the traces. One pair of two-microphone signals forms one intensity result. Therefore the number shall be even.

Mic. Spacing (mm): The spacing between two microphones. Default value is 16 mm.

Air Density (kg/m^ 3): Default value is 1.2.

Vertical Loads (kg): Test tire loads specified in **4.1.9**.

Comments (Defined as USER ID): Comments written here will be saved as USER ID associated with the calculated intensity for future reference or searching. Usually, the information given here includes tire manufacturer, tire name, and tire model, etc.

Time History Length (seconds): Default value is 15.65 seconds that the FFT algorithm uses to calculate a 32 Hz spectrum with 50% overlapping and 1000 average.

Scale w.r.t. Calibration (dB): Mainly used to compensate for the scale difference that often occurs in road tests. To maximize the signal/noise ratio, the microphone amplifier and date tape recorder are set for different dB scales when measuring road inputs with respect to calibration pure tone inputs. Their dB scale difference shall be input here. If the time traces for road inputs are correctly adjusted for the scale difference, “zero” dB shall be input here. Default value is zero.

Source BDM to Store Intensity: Specify the BDM (Block Data Memory – RAM Address used in LMS measurement system) here. The calculated intensity will be saved in the BDM for display.

Leading+Trailing, Average, All Toggle Buttons: The choice of the calculated intensity results. **Leading+Trailing** means to calculate the leading and trailing intensities; **Average** means to calculate the average of leading and trailing intensities; **All** means to calculate the leading and trailing intensity and their average. Default is to choose **Average**.

Road and **Dyno** Toggle Buttons: Specify the input data from either road tests or dynamometer tests for storing information. Default is to choose **Dyno**.

No Coherence Check and **Coherence Check** Toggle Buttons: Not activated yet.

The UPA program for calculating 32 Hz narrow-band intensity spectrums can be found in the TMON NVC UPA Programs.

Appendix E: Photos/Figures

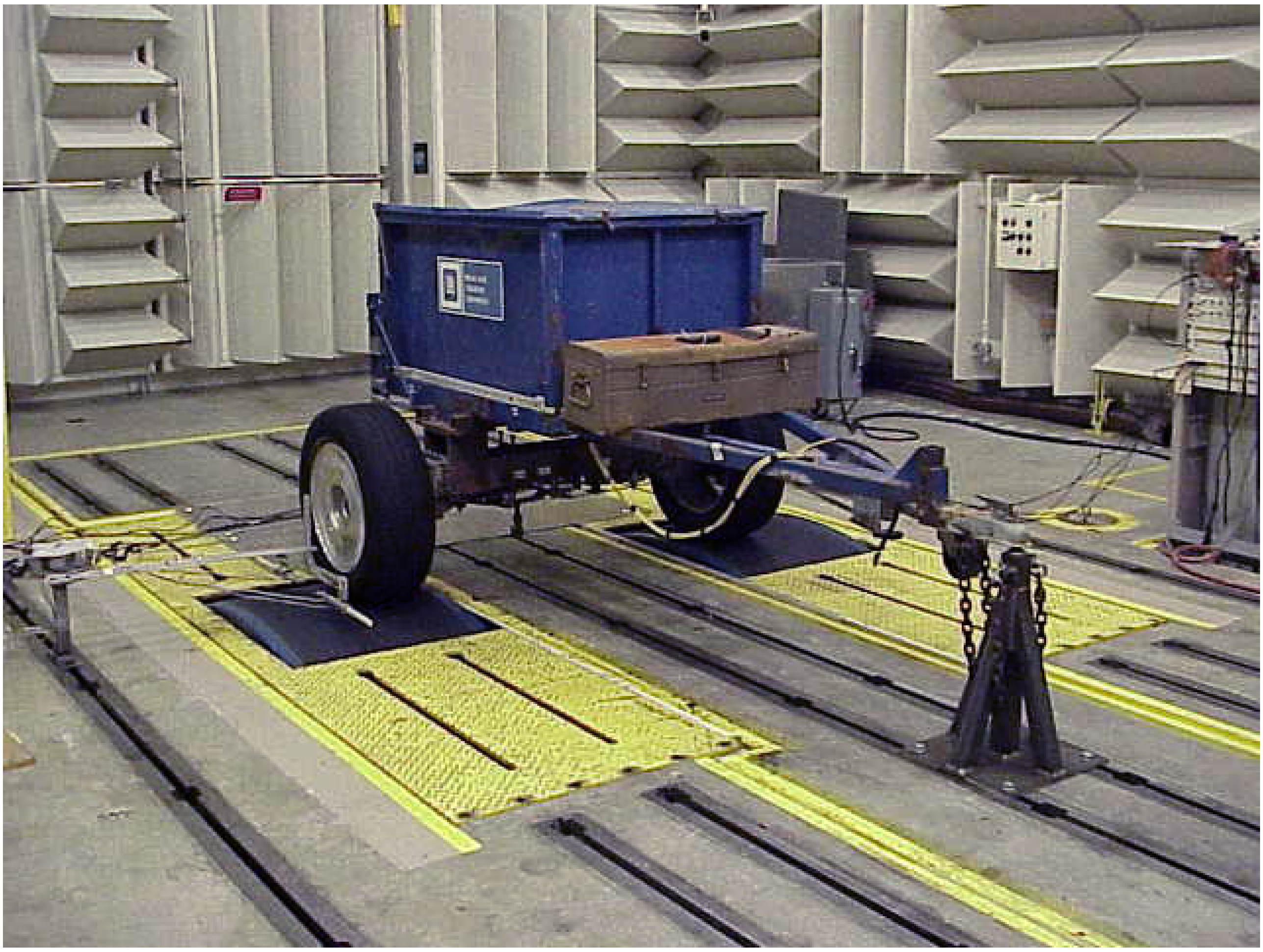


Figure E1: Trailer Setup in Hemi-Anechoic Dynamometer Chamber

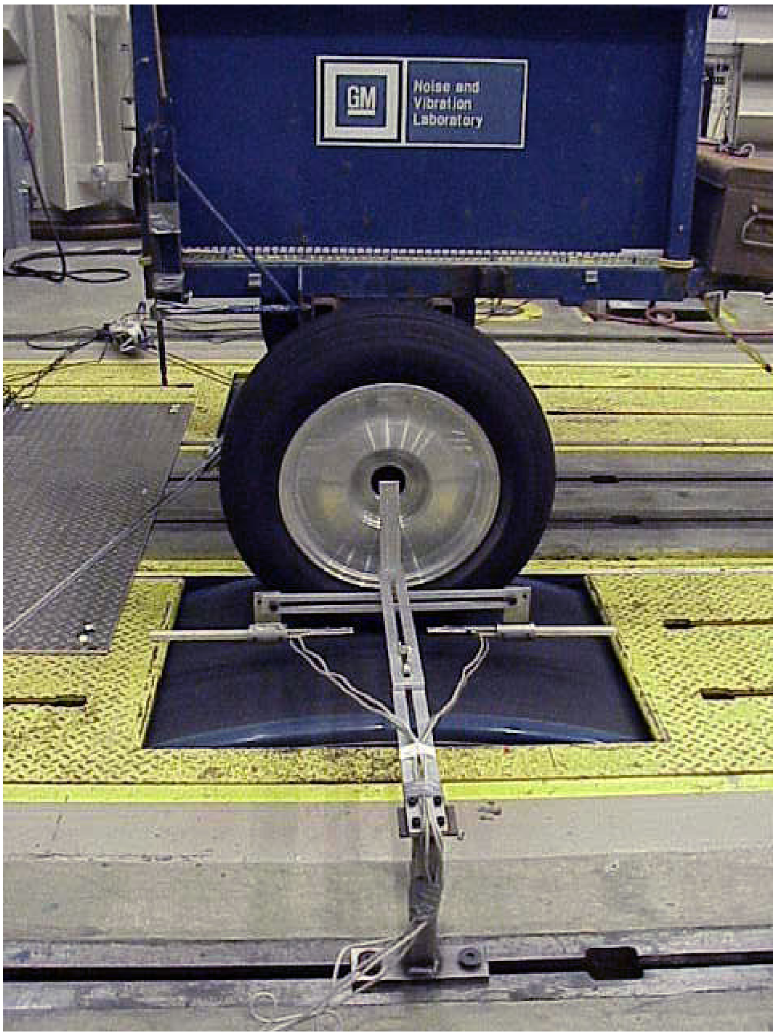


Figure E2: Fixture Setup with Intensity Probe Mounted

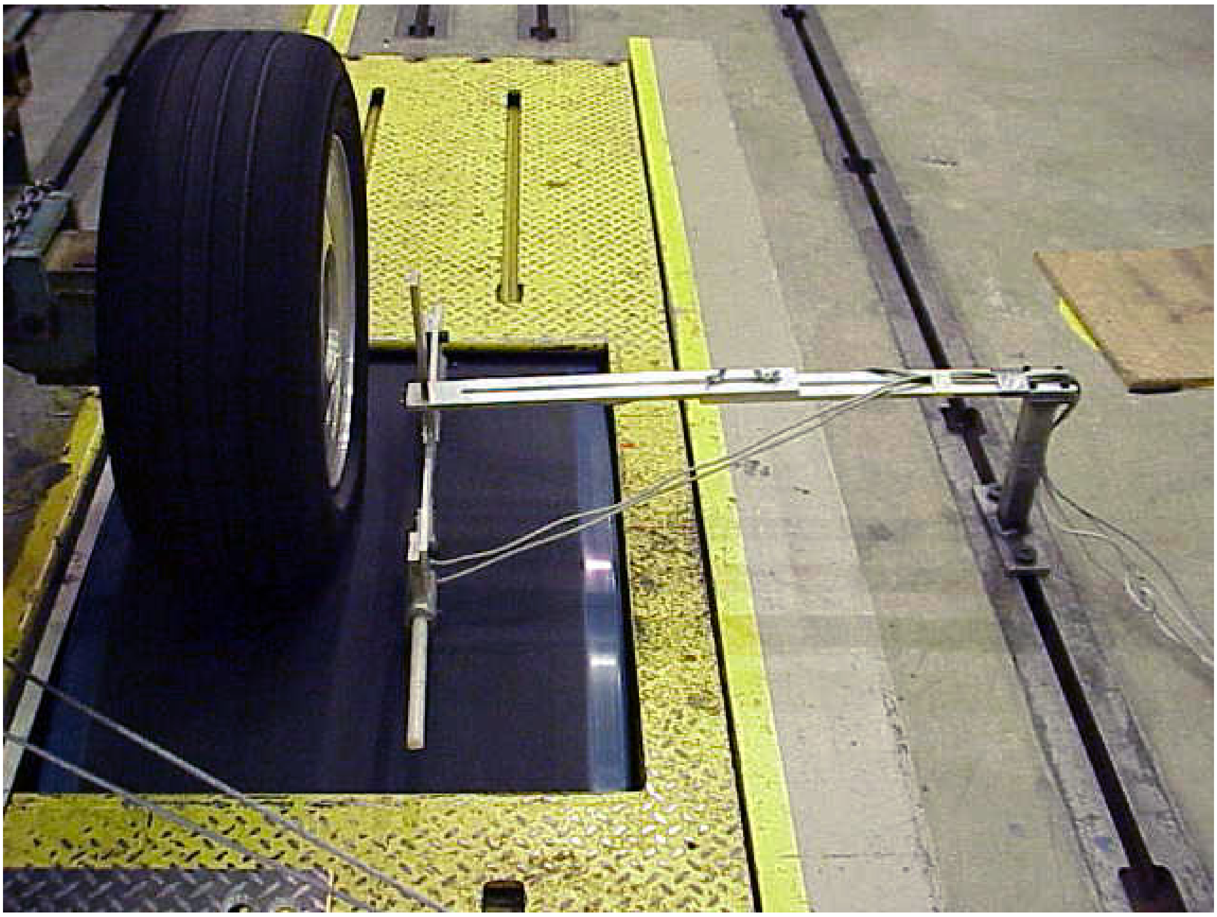


Figure E3: Fixture Setup with Intensity Probe Mounted

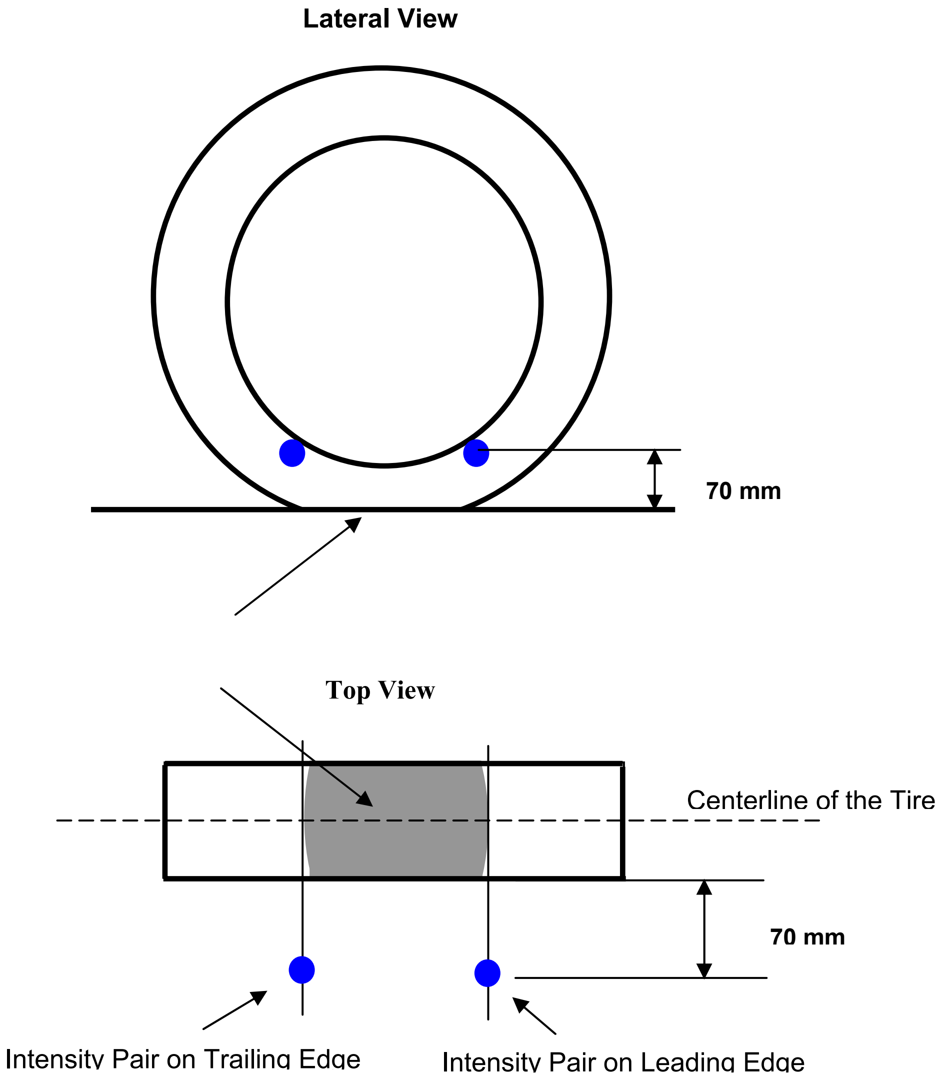
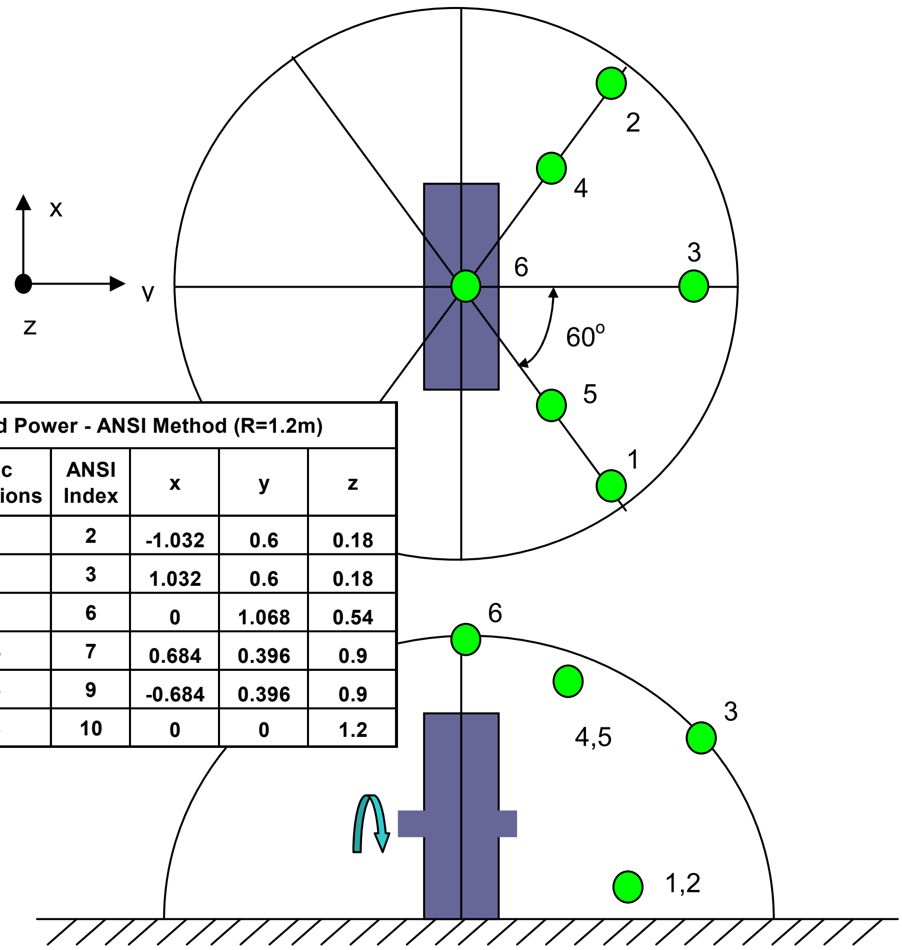


Figure E4: Schematic of Intensity Pair on Both Leading and Trailing Edges



Sound Power - ANSI Method (R=1.2m)				
Mic Positions	**ANSI Index**	**x**	**y**	**z**
1	2	-1.032	0.6	0.18
2	3	1.032	0.6	0.18
3	6	0	1.068	0.54
4	7	0.684	0.396	0.9
5	9	-0.684	0.396	0.9
6	10	0	0	1.2

Figure E5: Schematic of Sound Power Measurement in Accordance with ANSI S12.35-1990

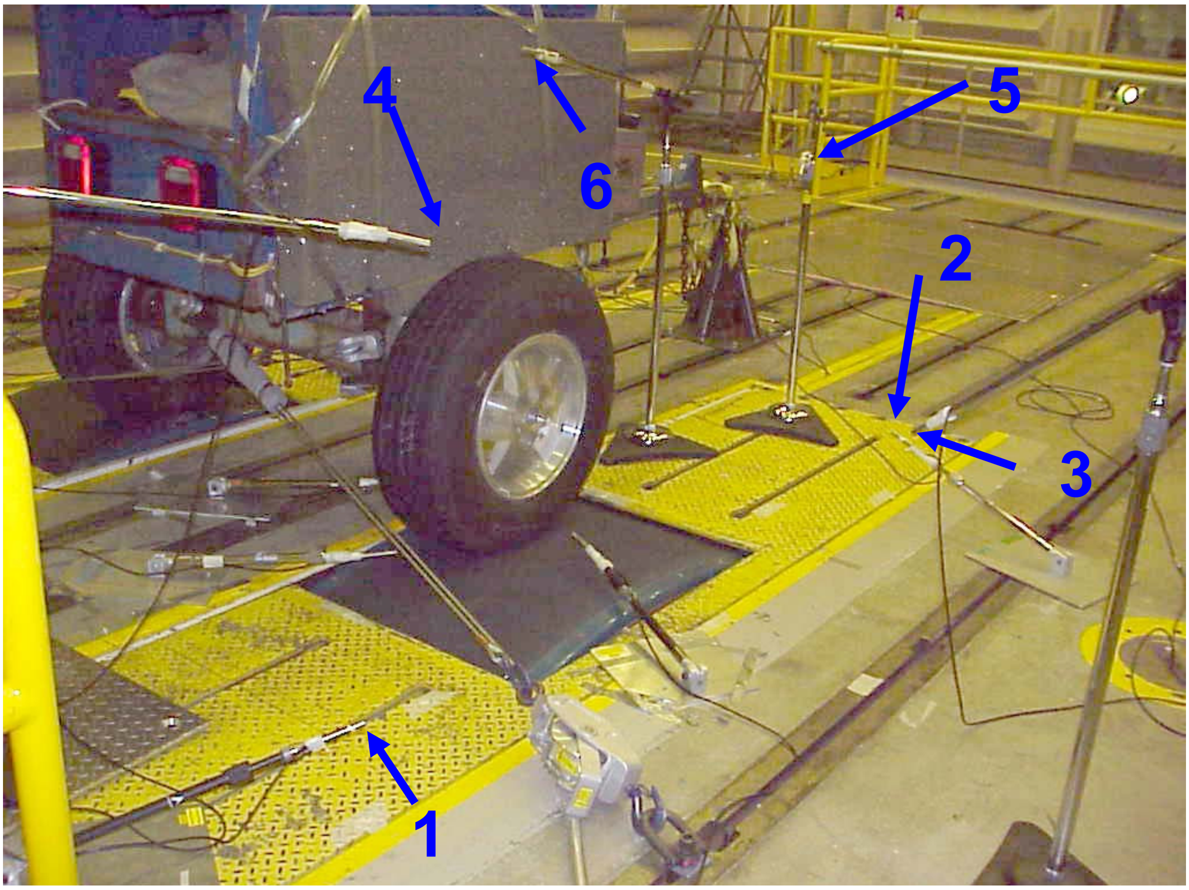


Figure E6: Six Microphone Positions for Far-Field Sound Power Measurement